| Problem Chosen | 2026 MCM/ICM Summary Sheet | Team Control Number |
|---|---|---|
| B | | 2631051 |

# Flying to the Moon: the Space Elevator with Rockets optimizing Planetary-Scale Transport System


## Summary

To support the establishment of a sustainable lunar colony for 100,000 people from 2050, this study develops and compares three Earth-to-Moon transportation strategies: a dedicated space elevator system, existing rocket launch infrastructure, and a hybrid system combining both. A multi-objective optimization model is formulated to minimize total cost and total transportation time, incorporating key parameters such as payload capacity, launch frequency, energy consumption, and system reliability. Monte Carlo simulations are integrated to assess the impacts of potential system failures, and environmental impacts—primarily carbon emissions—are quantified to evaluate sustainability.

**Problem 1 Comparative Analysis of Transport Strategies:** A bi-objective optimization model is developed to minimize total cost and time for transporting 10 metric tons of lunar colony materials. Three strategies are evaluated: space elevator only, rocket only, and a hybrid system. The space elevator alone requires 186 years, while rockets shorten the timeline to 164 years at significantly higher cost. The hybrid model provides an optimal trade-off, completing transport in 87.1 years at a cost of 18.2 trillion USD, with 46.75% of mass assigned to the elevator and the rest to rockets.

**Problem 2 Risk-Aware Transport Under System Failures:** Using Monte Carlo simulation, failure risks for both elevator components and rocket launches are incorporated into the hybrid model. Assuming a 3% launch failure probability, the optimized risk-adjusted allocation shifts to 52.3% via elevator and 47.7% via rockets, ensuring robust planning under uncertainty.

**Problem 3 Water Supply Logistics Optimization:** An EOQ-based inventory-replenishment model is formulated to minimize the annual cost of supplying water for 100,000 colonists.And the daily water consumption is set to three gears to find the closest to the actual water supply demand. The model determines optimal shipment quantities and replenishment cycles under elevator, rocket, and hybrid transport modes, balancing transportation and the Moon storage costs.

**Problem 4 Environmental Impact Integration:** A tri-objective optimization framework is introduced to minimize cost, time, and environmental impact—measured primarily through CO emissions. Strategies including carbon taxation, emission caps, supply of resources and clean transport prioritization are analyzed. Results show that an 85% reduction in emissions is achievable with a 15% cost increase and 17% longer timeline compared to the cost-optimal case.

We recommend a flexible, multi-modal transport plan that prioritizes the elevator for baseline supply and uses rockets for time-critical deliveries. This approach balances cost, schedule, uncontrollability,supply sustainability and environmental responsibility, providing a robust decision-making framework for large-scale lunar colonization logistics.



**Keywords:** Monte Carlo simulation, environmental impact, logistics planningMulti-objective optimization, Hybrid transport strategy, Risk-aware logistics planning, Sustainable lunar colonization, Carbon-emission minimization

# Contents


1 Introduction . . . 2
1.1 Problem Background . . . 2
1.2 Restatement of the Problem . . . 2
1.3 Literature Review . . . 2
1.4 Our Work . . . 4
2 Assumptions . . . 4
3 Notations . . . 6
4 Model Preparation . . . 6
4.1 Data Sources Overview . . . 7
4.2 Key Space Elevator Construction Parameters . . . 7
4.3 Modeling Assumptions . . . 7
4.4 Summary . . . 7
5 Models and Solutions of Problem 1 . . . 7
5.1 Models and Solutions of Problem a . . . 7
5.2 Models and Solutions of Problem b . . . 8
5.3 Models and Solutions of Problem c . . . 10
5.4 Comparison . . . 11
6 Models and Solutions of Problem 2 . . . 11
6.1 Space Elevator System Failure Modeling . . . 11
6.2 Rocket System Failure Modeling . . . 12
6.3 Mixed Transportation Optimization Under Failure Risk . . . 13
6.3.1 Parameters . . . 13
6.3.2 Objective 1: Minimize Total Cost . . . 14
6.3.3 Objective 2: Minimize Total Time . . . 14
6.3.4 Numerical Results . . . 14
7 Models and Solutions of Problem 3 . . . 15
7.1 Model Establishment . . . 15
7.2 Water Demand Calculation . . . 15
7.2.1 Inventory-Replenishment Optimization Model (EOQ Variant) . . . 15
7.2.2 Transportation Cost . . . 16
7.2.3 Storage Cost . . . 16
7.3 Constraints . . . 16
8 Models and Solutions of Problem 4 . . . 17
8.1 Model Assumptions . . . 17
8.2 Data Processing . . . 18
8.3 Model Formulation . . . 18
8.3.1 Multi-Objective Optimization Model . . . 19
8.4 Environmental Optimization Strategies . . . 20
8.4.1 Strategy 1: Environmental Damage Monetization . . . 20
8.4.2 Strategy 2: Environmental Constraint . . . 20
8.4.3 Strategy 3: Clean Transport Priority . . . 20
9 Sensitivity Analysis . . . 20
9.1 Electricity Price Ratio – Energy Consumption – Maintenance Coefficient . . . 20
9.2 Single Capacity and Single Launch Cost of Transport Rockets . . . 21
9.3 Unit Storage Cost . . . 21
10 Strengths and weaknesses . . . 21
11 Conclusion . . . 22

# 1 Introduction

## 1.1 Problem Background

The Moon Colony Management (MCM) aims to establish a Moon colony with a capacity of 100,000 people by 2050. This project will rely on an electrically powered space elevator system, along with existing Earth-based rocket launch facilities, to transport construction materials and supplies for the Moon colony and its inhabitants. In the proposed plan, three Galactic Harbours of the space elevator system are located in Ecuador (South America), Gabon (Africa), and Indonesia (Southeast Asia). These sites are positioned near the equator and are approximately 120 degrees apart from each other, with each connected to an Earth port. To ensure the smooth operation of the project, the MCM has tasked us with determining the optimal solution that minimizes both transportation costs and time expenditures.

## 1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Problem 1**
  For the transportation plan of the 100 million units of materials required to construct a Moon colony capable of accommodating 100,000 people, it is necessary to compare three approaches: relying solely on the space elevator system, existing Earth-based rocket launch bases, or conbining the both methods, in order to identify the optimal solution.

- **Problem 2**
  When the transportation system encounters failures such as cable oscillation, rocket failure, or elevator malfunction, these plans require adjustments to minimize losses in both time and cost.

- **Problem 3**
  To address the annual water supply needs for 100,000 residents in the Moon colony and ensure sustainable and adequate water provision once fully operational, the additional costs and timeline required will be explored using the delivery model.

- **Problem 4**
  This study will examine the environmental impact on Earth resulting from constructing a Moon colony capable of agricultural self-sufficiency, and analyze strategies to mitigate such impacts.

## 1.3 Literature Review

Research on large-scale Earth–Moon cargo transportation has primarily focused on three modes: the space elevator, reusable rockets, and hybrid systems combining both. Each strand of literature provides foundational insights into technical feasibility, operational economics, risk, and environmental impact, which collectively inform our modeling approach.

Technical studies on space elevators concentrate on structural dynamics, material science, and system reliability. Luo et al. (2022) conducted a series of analyses on the tether structure, climber dynamics, and stability of a segmented elevator, offering a theoretical basis for our structural assumptions

[2]. For system reliability, NASA standards such as NASA-STD-8729.1A provide a framework for designing large-scale repairable systems, which we adopt in our failure modeling [3, 4]. Energy supply remains a critical challenge; research by JAXA on microwave-based wireless power transmission offers a potential solution for powering climbers, supporting our assumption of clean energy operation [12]. The International Space Elevator Consortium (ISEC) regularly publishes updated dynamics and operational studies, which inform our capacity and throughput parameters [1].

Reusable rocket technology has dramatically reduced launch costs. Data from SpaceX indicate that refurbishment and reflight can lower expenses significantly compared to expendable launches [13]. Historical launch data also provide failure probability estimates ($\sim 3\%$), which we integrate into our risk model [9, 13]. Cost models for reusable rockets often distinguish between first-flight and subsequent-flight costs, a structure reflected in our transportation cost functions.

Given the complementary advantages of elevators (high capacity, low emissions) and rockets (speed, flexibility), recent studies have focused on optimizing hybrid systems. A common methodology involves multi-objective optimization minimizing cost and time, often solved via Pareto frontier analysis. Our model extends this approach by incorporating mass allocation between the two modes as a decision variable, similar to frameworks seen in integrated space logistics studies.

The environmental footprint of space transportation is increasingly scrutinized. Rockets emit significant $CO_2$—approximately 55 tons per ton of cargo to the Moon—whereas elevators powered by clean energy have near-zero operational emissions [11]. Carbon pricing mechanisms, such as those discussed by the U.S. EPA, offer a way to internalize environmental costs [6]. The Intergovernmental Panel on Climate Change (IPCC) reports on remaining carbon budgets inform our emission cap setting [5]. These sources collectively support our tri-objective optimization incorporating cost, time, and emissions.

For sustained lunar operations, resource replenishment models are essential. NASA's technical briefs on water systems and life support inform our water demand estimates [7]. Economic Order Quantity (EOQ) models, widely used in terrestrial supply chain optimization, are adapted here for lunar conditions, balancing shipment frequency against storage costs—an approach supported by space logistics literature.

Current literature often relies on theoretical or projected parameters, with limited integration of dynamic failure risks, environmental costs, and large-scale inventory planning under uncertainty. Our work addresses these gaps by:

- Combining multi-objective optimization with Monte Carlo simulation to model system failures and reliability.
- Incorporating carbon emissions and policy instruments (taxes, caps) into the optimization framework.
- Extending EOQ-based inventory models to the lunar context with hybrid transportation modes.
- Providing a comparative analysis of three transport strategies under unified cost, time, risk, and environmental metrics.

This integrated approach offers a more comprehensive decision-support tool for lunar colonization logistics, bridging engineering, economics, and environmental stewardship.

## 1.4 Our Work

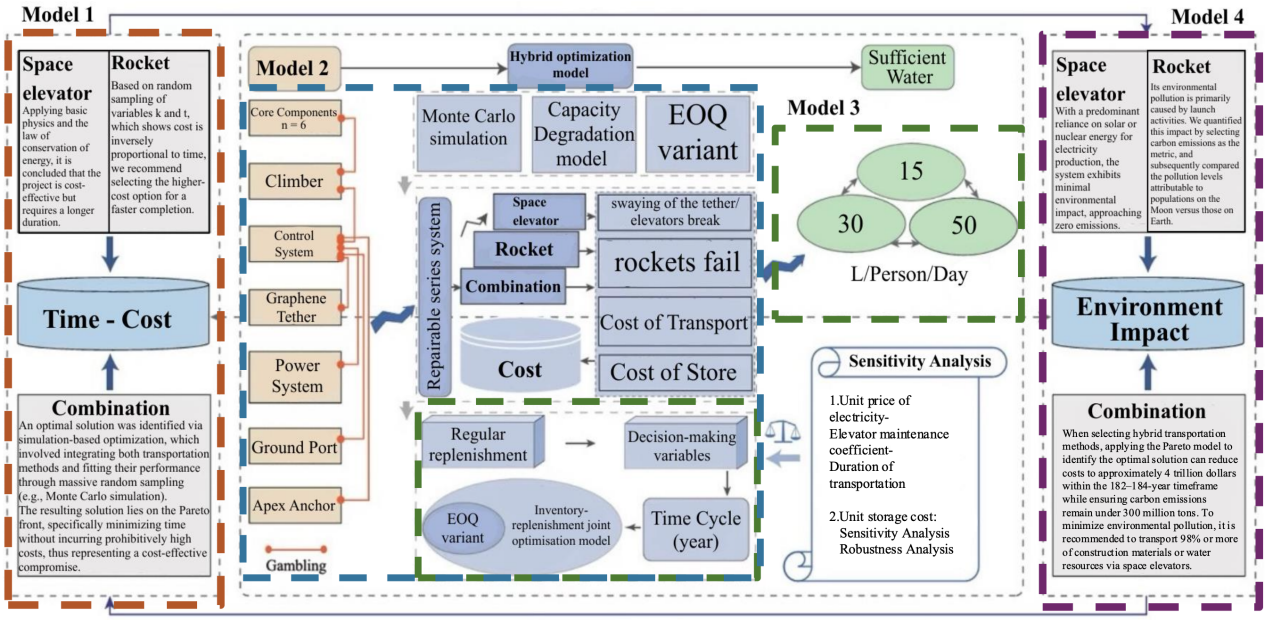



Figure 1: Mind Map

# 2 Assumptions

Before establishing a mathematical model for the space elevator system and rocket launch bases employed for the construction and maintenance of the lunar colony. we make some assumptions to make the model easier to realize.

- **Assumption 1: Neglecting the errors arising from the separate locations of the Galactic Harbours and the rocket launch sites.**
  This assumption treats different launch sites on Earth as equivalent in terms of energy requirements, thereby simplifying the positional variables in orbital mechanics calculations. In reality, factors such as launch latitude, Earth's rotational speed, and atmospheric conditions affect launch efficiency. However, this assumption aims to establish a location-independent theoretical energy baseline, allowing the model to focus on the overall optimization of interstellar transport routes rather than local geographical differences.

- **Assumption 2: Under identical distances from Earth, rockets maintain the same launch velocity.**
  This assumption simplifies the launch model by treating the required launch velocity as the same from equal distances to Earth. It is based on the fixed energy needed to reach a specific orbit, ignoring real-world factors like Earth's rotation and atmospheric drag. This provides a theoretical baseline for comparing mission energy costs before adding practical adjustments.

- **Assumption 2: Different material transport events within the model are independent occurrences and do not influence one another.**

This assumption models the system as an idealized, congestion-free channel with instant resource availability. Consequently, total cost and time become linear sums of individual transport events. Each event functions as an independent computational unit with deterministic inputs and outputs. This eliminates the uncertainties and randomness inherent in concurrent operations. It allows the analysis to focus on the impact of core technical parameters.

- **Assumption 3: The duration of rocket launch is neglected.**
  The duration is too short compared to the entire transport process and can be neglected.

- **Assumption 4: Throughout the construction and maintenance period of the lunar colony, the initial semi-major axis of the lunar orbit is approximately $a \approx 3.847 \times 10^8$ m with an eccentricity of $e \approx 0.055$.**
  Although the lunar orbit experiences long-term perturbations and periodic variations, its orbital parameters change only slightly within the mission time scale (e.g., several years). Fixing the orbital parameters simplifies the design and energy calculations of Earth-Moon transfer trajectories, providing a stable astronomical environment baseline for transport costs and avoiding the complexity and uncertainty introduced by orbital oscillations.

- **Assumption 5: The rocket launch pads are lossless and capable of supporting multiple repeated launches.**
  This assumption treats launch pads as idealized infrastructure, ignoring real-world limitations such as maintenance, wear, and queueing times. It enables the model to focus on the energy and time costs of rocket transport itself rather than operational constraints of ground facilities. Consequently, the impact of core technical parameters on transport efficiency can be better evaluated at the system level, consistent with the overall modeling principle of "independent events with linear superposition."

# 3 Notations

Table 1: Notation Summary

| Group 1 | | | Group 2 | | |
|---|---|---|---|---|---|
| Symbol | Description | Unit | Symbol | Description | Unit |
| $M$ | Total mass | ton | $x$ | Mass to elevator | ton |
| $y$ | Mass to elevator (risk) | ton | $a$ | Semi-major axis | m |
| $e$ | Eccentricity | – | $R$ | Earth-Moon distance | m |
| $R_e$ | Earth radius | m | $\mu$ | Grav. parameter | $m^3/s^2$ |
| $C$ | Cost | USD | $c$ | Unit cost | USD/ton |
| $c_e$ | Elevator unit cost | USD/ton | $c_o$ | Orbital rocket cost | USD/launch |
| $C_{\text{total}}$ | Total cost | USD | $C_e$ | Elevator cost | USD |
| $C_r$ | Rocket cost | USD | $C_o$ | Orbital rocket cost | USD |
| $C_{\text{transport}}$ | Transp. cost/shipment | USD | $C_{\text{storage}}$ | Storage cost | USD |
| $C_{\text{env}}$ | Environ. cost | USD | $P$ | Power/capacity | W/ton/launch |
| $P_{\text{th}}$ | Single port capacity | ton/yr | $P_{\text{atotal}}$ | Total elevator cap. | ton/yr |
| $P_r$ | Rocket payload | ton/launch | $n$ | Launch count | count |
| $n_r$ | Rocket launches | count | $n_o$ | Orbital launches | count |
| $T$ | Time | day/yr | $T_{\text{total}}$ | Total time | yr |
| $T_e$ | Elevator time | yr | $T_r$ | Rocket time | yr |
| $t_0$ | Flight time (Hohmann) | day | $t_{\text{rep}}$ | Repair time | day |
| $g$ | Grav. acceleration | $m/s^2$ | $h$ | Height/storage cost | m/USD/ton/yr |
| $\eta$ | Efficiency | – | $k$ | Coefficient | –/count |
| $k_m$ | Maint. coeff. | – | $p$ | Failure prob. | – |
| $p_f$ | Rocket fail prob. | – | $A$ | Availability | – |
| $A_i$ | Comp. avail. | – | $A_{\text{total}}$ | Sys. avail. | – |
| $N_s$ | Launch sites | count | $N_{\text{suc}}$ | Success launches | count |
| $N_{\text{total}}$ | Total launches | count | $N_{\text{fail}}$ | Failed launches | count |
| MTBF | Mean time between failures | day | MTTR | Mean time to repair | day |
| $D$ | Annual demand | ton/yr | $d$ | Daily consumption | L/pers/day |
| $Q$ | Shipment quantity | ton | $s$ | Replen. cycle | yr |
| $R_e$ | Elevator rate | ton/yr/trip | $R_r$ | Rocket rate | ton/yr/launch |
| $w$ | Mass to elevator (replen.) | ton | $e_e$ | Elevator CO coeff. | ton CO/ton |
| $e_{\text{launch}}$ | Rocket CO coeff. | ton CO/ton | $e_{\text{tran}}$ | Transfer rocket CO coeff. | ton CO/ton |
| $E$ | Total environ. impact | ton CO | $p_c$ | Carbon tax rate | USD/ton CO |
| $B$ | CO cap | ton CO | $N$ | Population | pers |
| $\rho$ | Density (e.g., water) | kg/L | | | |

Note: r = Rocket; e = Elevator.

# 4 Model Preparation

To construct a robust Earth–Moon transport optimization model, we prepared foundational physical parameters, engineering assumptions, and external datasets. This section summarizes the data sources, parameter origins, and physical relations supporting subsequent analyses.

## 4.1 Data Sources Overview

**Space elevator:** segmented tether structure and dynamics; tether/climber interface; energy and wireless power transmission; maintenance cost proportional to electricity cost (typical range 0.18–0.22).
**Rockets:** reusable launch pricing (first vs. reused), turnaround (1–2 days), Hohmann transfer time (∼4.98 days), and a representative failure probability (∼3%).
**Environment:** rocket emission coefficients (e.g., ∼55 t $CO_2$ per ton to the Moon) vs. near-zero elevator emissions under clean power; carbon pricing and emission caps for tri-objective optimization.
**Astrodynamics:** Earth radius $R_e$, Earth gravitational parameter $\mu$, lunar mean distance $R$, and eccentricity $e$ as adopted in the orbital calculations.

## 4.2 Key Space Elevator Construction Parameters

**Tether materials:** graphene/CNT composite bands with high specific strength and safety factors compatible with segmented elevator analysis; climber–tether interface per ISEC; apex-anchor and ground-port components included in availability modeling.
**Annual throughput:** $P_{\text{harbour}} = 179{,}000$ tons/year per harbour; three harbours yield $P_{\text{total}} = 537{,}000$ tons/year.
**Power requirement:** derived from energy conservation $E \approx mgh/\eta$ with system efficiency $\eta$ and a maintenance coefficient $k_m \approx 0.20$ (ESA guideline median).

## 4.3 Modeling Assumptions

We assume stable lunar orbital parameters over the mission horizon, independence among transport events, site equivalence for launch bases, a constant launch failure probability for planning purposes, and near-zero elevator emissions under clean energy. These assumptions simplify tractable optimization while aligning with the cited literature.

## 4.4 Summary

Our preparation integrates orbital mechanics, structural constraints, energy and maintenance models, launch economics, statistical reliability, and environmental coefficients. This foundation supports bi-objective (cost–time) optimization, risk-aware mixed transport planning, EOQ-based resupply modeling, and tri-objective environmental trade-offs used throughout the study.

# 5 Models and Solutions of Problem 1

## 5.1 Models and Solutions of Problem a

From the problem statement, the capacity of a single space elevator, denoted $P_{\text{harbour}}$, is 179,000 metric tons per year. The Space Elevator System comprises three Galactic Harbours. Therefore, the total system capacity $P_{\text{a total}}$ is:

$$P_{\text{a total}} = 3P_{\text{harbour}} = 3 \times 179{,}000 = 537{,}000 \text{ metric tons per year.} \quad (1)$$

It is estimated that the Moon Colony will necessitate approximately M = $1 \times 10^8$ metric tons of materials. Therefore, the total system durition $T_{\text{a total}}$ is:

$$T_{\text{a total}} = \frac{M}{P_{\text{a total}}} \approx 186.22 \text{ years} \tag{2}$$

The costs of the space elevator system can be divided into two main categories: the unit cost of the elevator and the cost of orbital rockets.

$$C_{\text{a total}} = C_{\text{e}} + C_{\text{o}} \tag{3}$$

The unit cost of the elevator is defined as the unit electricity cost plus the unit maintenance cost.Based on the law of conservation of energy, the unit electricity cost can be derived as:

Assuming that the space elevator is operating normally, the single maintenance cost is attributed to regular inspections.Hence, it can be approximated as a constant value.

The European Space Agency (ESA) official website, "Space Infrastructure Maintenance Guidelines (2024 Edition)," states: "For large-scale space transportation systems (such as orbital elevators and space station cargo channels), maintenance costs typically amount to 0.18-0.22 of electricity costs." Using the median value of 0.20 aligns with the design requirements of a space elevator for "high reliability and low failure rates."

$$\begin{cases} C_e = c_e \cdot M = C_p + C_m \\ C_m = C_p \cdot k_m, \quad k_m = 0.2 \\ C_p = \dfrac{1000 m_0 \cdot g \cdot h \cdot P}{\eta \cdot 3.6 \times 10^6} \end{cases} \tag{4}$$

Total cost of elevators:

$$C_e = 7.467 \times 10^4 m_0 USD \tag{5}$$

The starting point at 100,000 kilometers does not require overcoming ground gravity or atmospheric drag, and its payload capacity far exceeds the 150-ton limit of ground-based rockets. Number of launches:

$$n_o = \lceil M/P_o \rceil = 2 \times 10^5, \quad P_o = 500 \text{ ton/time} \tag{6}$$

Given:

$$M = 10^8 \text{ ton}, \quad P_o = 500 \text{ ton/launch}, \quad c_o = 2 \times 10^7 \text{ USD/launch}.$$

Total cost of orbital rockets:

$$C_o = n_o \cdot c_o = (2 \times 10^5) \times (2 \times 10^7) = 4 \times 10^{12} \text{ USD}.$$

Total cost of the system:

$$C_{\text{a total}} = C_{\text{e}} + C_{\text{o}} = 7.467 \times 10^{12} + 4 \times 10^{12} \approx 11.467 \times 10^{12} \text{USD} \tag{7}$$

## 5.2 Models and Solutions of Problem b

According to the problem, Option B will involve the selective use of ten existing rocket launch bases on Earth. Based on data from SpaceX, the turnaround time for rocket recovery and reuse is projected to be shortened to 1 to 2 days by 2050. In our model, we examine the transportation costs and scheduling under scenarios where the turnaround period is 1 day or 2 days, respectively.

The flight duration from Earth to the Moon via a Hohmann transfer orbit is computed considering an Earth-centered elliptical orbit with perigee near Earth's surface and apogee at the lunar orbit. The given parameters are: Earth-Moon distance $R = 3.847 \times 10^8$ m, lunar orbital eccentricity $e = 0.055$ (not directly used in the calculation, as the Hohmann transfer assumes a circular target orbit, with the mean distance $R$ taken as the orbital radius). Earth radius is taken as $R_e = 6.371 \times 10^6$ m, and Earth's gravitational parameter $\mu = 3.986 \times 10^{14}\ \mathrm{m^3\ s^{-2}}$.

The semi-major axis $a$ of the Hohmann transfer orbit is half the sum of the perigee and apogee radii:

$$a = \frac{R_e + R}{2} = \frac{6.371 \times 10^6\ \mathrm{m} + 3.847 \times 10^8\ \mathrm{m}}{2} = 1.955\,355 \times 10^8\ \mathrm{m}. \tag{8}$$

The flight duration is half the period of the transfer orbit:

$$t_0 = \pi \sqrt{\frac{a^3}{\mu}}. \tag{9}$$

Thus, the flight duration is approximately $4.30 \times 10^5$ s, equivalent to about 4.98 days.

**Note:** The calculation neglects launch and landing times and assumes a two-body problem with the lunar orbit approximated as circular. The Hohmann transfer is based on the mean distance. Considering the eccentricity of the lunar orbit would introduce minor variations, but the present result suffices for the required precision based on the given $R$.

From the problem, it can be obtained that:

$$\begin{cases} P_r = 150\ \text{t/time}, \\ n_r = \lceil M/P_r \rceil, \\ T_r = t_0 \cdot n_r/k + t \cdot (n_r/k - 1), \\ t_0 = 4.98 days \end{cases} \tag{10}$$

The total transportation time of the rocket is:

$$T_r = \frac{6.7 \times 10^5}{k}(4.98 + t) - t \tag{11}$$

The cost analysis accounted for both rocket launch and maintenance expenditures. For launch costs, the procurement fees of the SpaceX Falcon rocket were applied, with an initial launch cost of 90 million USD and an innovative reduced cost of 36 million USD for reused vehicles. Maintenance costs, covering equipment upgrades, personnel allocation, rapid refueling, and airframe reuse optimization, were modeled accordingly.

$$\begin{cases} C_r = c_f \cdot k + (n_r - k) \cdot c_b + C_u, \\ C_u(k,t) = ak + b/t, \\ C_f = 9.0 \times 10^7\ \text{USD}, \\ C_u = 3.6 \times 10^7, \\ M = 10^8 ton \end{cases} \tag{12}$$

The total transportation cost of the rocket is:

$$C_r = 1.9 \times 10^8\, k + 3.6 \times 10^7 \cdot (6.7 \times 10^5 - k) + \frac{10^9}{t} \tag{13}$$

Based on the aforementioned model, the fitting results (as shown in the Figure 2) indicate an inverse relationship between time and cost. When prioritizing time, utilizing existing rocket launch sites requires a minimum of 163.6 years at an approximate cost of 24 trillion USD. For a more specific solution, we can perform further fitting to provide a relatively optimal transportation cost and timeline plan for your consideration.

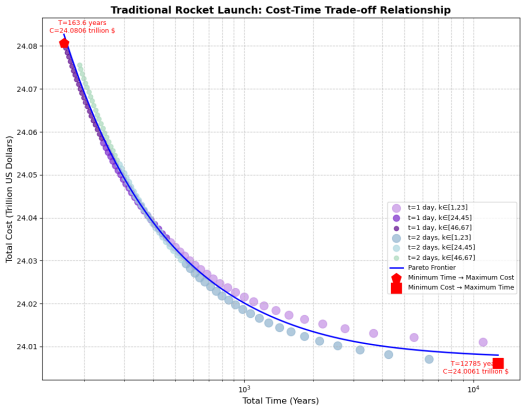



Figure 2: Rocket transport time - cost curve

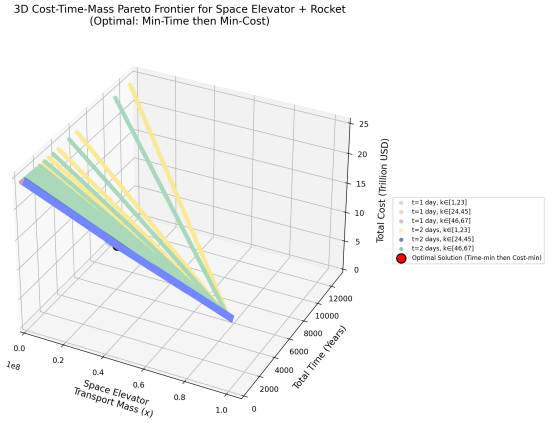



Figure 3: Three-dimensional diagram of hybrid transportation optimisation

## 5.3 Models and Solutions of Problem c

### Mixed Transportation Optimization Model

We consider a dual-mode transportation system for shipping cargo from Earth to space. The total mass to be transported is $M$ (tons). The decision variable $x$ denotes the mass assigned to the space elevator, while the remainder $M - x$ is transported by reusable rockets.

- Space elevator: $c_e$ – unit cost (\$/ton), $R_e$ – transport rate (tons/year).
- Rocket: $P_r$ – payload capacity (tons/launch), $C_1$ – first launch cost (\$), $C_2$ – subsequent launch cost (\$), with $C_2 < C_1$, $T_c$ – time per launch (years/launch).
- $T_{\max}$ – maximum allowed project duration (years).

Based on the aforementioned variables, the following model was designed to seek the optimal solution, which minimizes both time and cost.

Constraints:

$$0 \le x \le M, \qquad x \le R_e\, T_{\max}.$$

Number of rocket launches:

$$n_r = \left\lceil \frac{M - x}{P_r} \right\rceil, \quad n_r = 0 \text{ if } M - x = 0.$$

**Objective 1: Minimize total cost**

$$\min \ C_{\text{total}}(x) = c_e x + C_f \cdot k + \frac{M - x}{n_r - k} C_b + C_u$$

**Objective 2: Minimize total time**

$$\min \ T_{\text{total}}(x) = \max\left\{\frac{x}{R_e}, \ \frac{M - x}{P_r}\right\}.$$

This bi-objective optimization problem can be analyzed via Pareto frontiers, visualized in a three-dimensional space $(x, C_{\text{total}}, T_{\text{total}})$.

As can be visually identified from the Figure 3, the red point marks the optimal solution, requiring approximately 87.1 years and a cost of about 18.2 trillion USD. In this scenario, space elevators handle 46.75% of the total mission mass M. The plan necessitates the deployment of all 67 rocket launch pads with an assumed launch interval of 24 hours.

## 5.4 Comparison

If your primary objective is the fastest completion, it advises selecting Option c (from Problem 1). Conversely, if maximizing profitability (minimizing cost) is the key driver, then Option a (187-year timeline) is optimal. For further comparison, please refer to the time-cost chart provided for Option b.

# 6 Models and Solutions of Problem 2

Monte Carlo simulation is well-suited for modeling low-probability, high-risk events due to its strengths in handling multidimensional uncertainties, capturing rare events, integrating multiple risk sources, and generating comprehensive risk metrics. This makes it highly suitable for building multi-modal transportation failure risk assessment models.Therefore, we employ Monte Carlo simulation, integrated with the model established in Problem 1, to analyze potential failures.

## 6.1 Space Elevator System Failure Modeling

By applying the Monte Carlo simulation method to the transport cost and schedule model constructed for the normal operation of the space elevator in Problem 1, we will identify how transport costs and time will change in the event of elevator damage.

The space elevator is treated as a series repairable system, and the steady-state availability of individual components is calculated using the classical repairable system formula.

$$A_i = \frac{\text{MTBF}_i}{\text{MTBF}_i + \text{MTTR}_i} \tag{14}$$

Here, $A_i$ represents the availability of the $i$-th component, $\text{MTBF}_i$ denotes the mean time between failures (in days) of the $i$-th component, and $\text{MTTR}_i$ refers to the mean time to repair (in days) of the $i$-th component.

Since the failure of any core component in the space elevator system will lead to a complete operational shutdown, the overall system availability is given by the product of the availability of all components:

$$A_{\text{total}} = \prod_{i=1}^{n} A_i \tag{15}$$

Here, $n$ represents the number of core components in the system. Six core components are selected, including the graphene tether, climber, ground port, apex anchor, power system, and control system, and the Mean Time Between Failures (MTBF) and Mean Time to Repair (MTTR) vary for each component.Where $n = 6$ represents the number of core components: graphene tether, climber, ground port, apex anchor, power system, and control system. The numerical reasonable interval gap of Table 2 is too small, so there is no need to do sensitivity analysis.

Table 2: MTBF and MTTR for space elevator components

| Component | MTBF (days) | MTTR (days) |
|---|---|---|
| Graphene Tether | 3650 | 30 |
| Climber | 1825 | 15 |
| Ground Port | 7300 | 5 |
| Apex Anchor | 5475 | 20 |
| Power System | 3650 | 10 |
| Control System | 1825 | 7 |

The effective annual capacity of the elevator system under failure risk is:

$$C_e = C_{\text{rated}} \times A_{\text{total}} \quad \text{(tons/year)} \tag{16}$$

where $C_{\text{rated}} = 537,000$ tons/year. The transport time under failure conditions becomes:

$$T_e = \frac{x}{C_e} \quad \text{(years)} \tag{17}$$

The results are shown in the Figure 4 below.

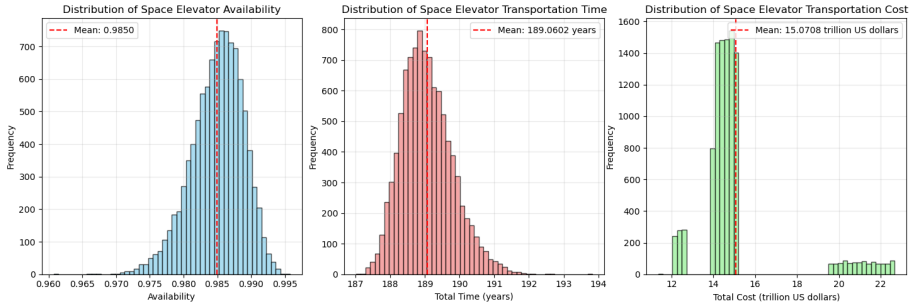



Figure 4: The results of Space Elevator

## 6.2 Rocket System Failure Modeling

It is assumed that the probability of launch failure for rockets is constant. The probability of different increases in time and costs due to failures will be determined through simulation methods. It

is assumed a constant launch failure probability $p_f = 0.03$ (3%) per launch based on historical SpaceX data. The number of successful launches required to transport mass $M - x$ is:

$$N_{\text{success}} = \frac{M - x}{P_r} \tag{18}$$

The total number of launches including failures follows a geometric distribution:

$$N_{\text{total}} = \frac{N_{\text{success}}}{1 - p_f} \tag{19}$$

The additional time due to failures includes repair and replacement time per failure event, assumed as $t_{\text{repair}} = 30$ days. The total time for rocket transport becomes:

$$T_r = N_{\text{success}} \cdot t_0 + (N_{\text{total}} - N_{\text{success}}) \cdot t_{\text{repair}} \tag{20}$$

where $t_0 = 4.98$ days is the nominal transit time per launch.

The cost impact includes failure recovery cost $C_{\text{fail}} = 5 \times 10^7$ USD per failure. Total rocket cost under failure risk:

$$C_r = C_{\text{normal}} + (N_{\text{total}} - N_{\text{success}}) \cdot C_{\text{fail}} \tag{21}$$

where $C_{\text{normal}}$ is the nominal cost from Problem 1, and the results are shown in the Figure 5.

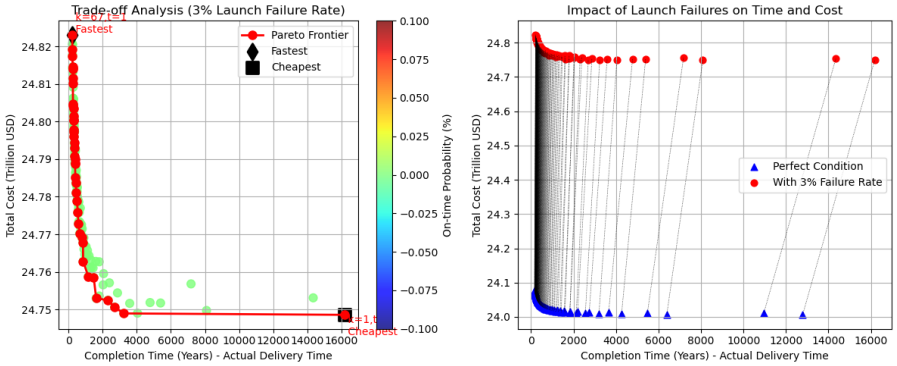



Figure 5: Rocket System Failure Modeling

## 6.3 Mixed Transportation Optimization Under Failure Risk

### 6.3.1 Parameters

- Total mass to transport: $M$ (tons)
- Decision variable: $x$ = mass assigned to space elevator (tons)
- Rocket transport mass: $M - x$ (tons)

We extend the mixed transportation model from Problem 1 to incorporate failure risks for both systems. Let $y$ be the mass assigned to the space elevator, and $M - y$ to rockets.

### 6.3.2 Objective 1: Minimize Total Cost

$$\min C_{\text{total}}(y) = C_e(y) + C_r(y) + C_{\text{env}}(y) \tag{22}$$

where:

$$C_e(y) = c_e \cdot y + C_{\text{maintenance}} \cdot \frac{y}{C_e}$$

$$C_r(y) = [F \cdot k + V \cdot (N_{\text{success}} - k)] \cdot \frac{1}{1 - p_f} + N_{\text{fail}} \cdot C_{\text{fail}}$$

$$N_{\text{fail}} = \left( \frac{1}{1 - p_f} - 1 \right) \cdot N_{\text{success}}$$

### 6.3.3 Objective 2: Minimize Total Time

$$\min T_{\text{total}}(y) = \max \left( \frac{y}{C_e \cdot A_{\text{total}}}, \frac{M - y}{P_r \cdot (1 - p_f)} + N_{\text{fail}} \cdot t_{\text{repair}} \right) \tag{23}$$

**Constraints**

$$0 \le x \le M, \quad x \le C_e \times T_{\max}$$

where $N_{\max}$ is the maximum allowable launches per year.

### 6.3.4 Numerical Results

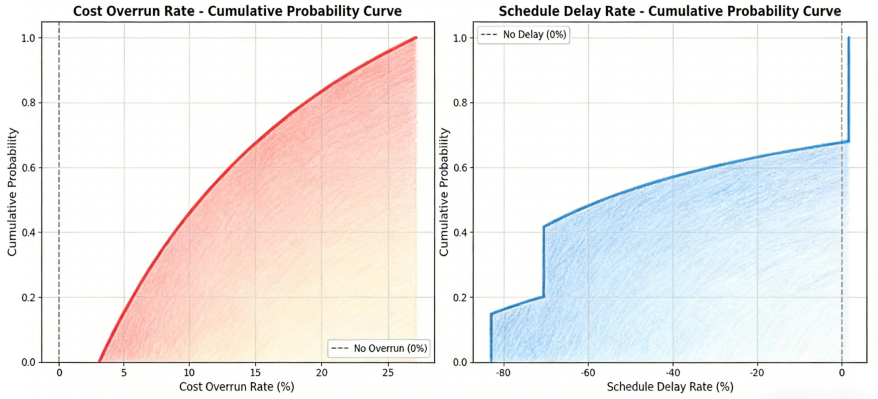



Figure 6: Numerical Results

Based on Monte Carlo simulation with 10,000 iterations, under a 3% failure rate:

- Average completion time increases by 28.6%
- Average cost increases by 3.1%
- Optimal allocation: 52.3% to elevator, 47.7% to rockets

This model integrates capacity degradation and reliability factors to provide realistic estimates for space transportation planning under uncertainty, enabling robust decision-making in the presence of system failures.

# 7 Models and Solutions of Problem 3

## 7.1 Model Establishment

When transporting water to off-Earth locations such as space stations in practice, iodine or silver ions are often added to enhance stability. However, due to the extremely small volume of these additives relative to water, their impact on overall density is negligible. Therefore, the density of pure water is adopted in this model for calculations and simulations.

Assuming that the machines and environment required for storing and recycling water are part of the basic infrastructure of the lunar colony.

This research is inspired by NASA engineering guidelines and space logistics studies. For the first fully operational year of the lunar colony, the total population is assumed to remain constant at 100,000, all of whom are prime working-age adults from Earth. During this period, both the birth rate and mortality rate are assumed to be zero.

This paper presents an optimization model for water supply logistics to lunar colonies, based on an Economic Order Quantity (EOQ) variant. The model balances transportation and storage costs to minimize the average annual total cost. We consider stable water demand, periodic replenishment strategies, and linear storage costs. Transportation systems include space elevators, rockets, and hybrid modes, with cost functions derived from fixed and variable components. The decision variables are the replenishment cycle and shipment quantity. Sensitivity analysis is conducted to assess robustness under varying parameters, such as per capita water consumption and storage cost rates. The model provides a framework for cost-effective water resource management in extraterrestrial settlements.

## 7.2 Water Demand Calculation

Let $d$ be the daily water consumption per capita in liters (L), with possible values of 15, 30, or 50 L/person/day, and water density $\rho = 1$ kg/L. For a colony of $N$ persons, the annual water demand $D$ in tons (t) is:

$$D = \frac{d \times N \times 365}{1000} \quad \text{t/year}, \tag{24}$$

where 1 L = 1 kg = 0.001 t. Demand is assumed constant over time, with no seasonal or fluctuations.

### 7.2.1 Inventory-Replenishment Optimization Model (EOQ Variant)

The model aims to minimize the average annual total cost, comprising transportation and storage costs. Decision variables are the replenishment cycle $s$ (years) and the shipment quantity per replenishment $Q$ (t), where $Q = D \times s$. The objective function is:

$$\min_{s,Q} \left( \frac{C_{\text{transport}}}{s} + \frac{Q}{2} \times h \right), \tag{25}$$

where $C_{\text{transport}}$ is the transportation cost per shipment (in USD), and $h$ is the unit storage cost (in USD/t/year). For a linear consumption pattern with periodic replenishment, the average inventory is $Q/2$.

### 7.2.2 Transportation Cost

Transportation cost depends on the system used:

1. **Space elevator system:**
$$C_{\text{transport}} = Q \times c_e, \tag{26}$$
where $c_e$ is the elevator unit cost (USD/t). This assumes linear cost with mass.

2. **Rocket system:** Let $R_r$ be the rocket's payload capacity (t/shipment), $F$ the first-launch cost per rocket (USD), $V$ the cost per launch after recovery (USD), and $E$ additional launch costs (e.g., for extra missions). For a shipment quantity $Q$, the number of launches is $\lceil Q/R_r \rceil$. Let $k$ be the number of first launches (e.g., new rockets), then the transportation cost is:
$$C_{\text{transport}} = F \times k + \left( \left\lceil \frac{Q}{R_r} \right\rceil - k \right) \times V + E. \tag{27}$$
Typically, $k$ is constrained by the available new rockets, but for simplicity, we assume $k$ is a decision variable or fixed based on mission plans.

3. **Hybrid system:** Let $w$ be the mass transported by elevator (t) in a shipment, so $Q - w$ is transported by rocket. The transportation cost combines both modes:
$$\begin{aligned} C_{\text{transport}} = w \times c_e + \left\lceil \frac{w}{R_e} \right\rceil \times C_e + F \times k \\ + \left( \left\lceil \frac{Q - w}{R_r} \right\rceil - k \right) \times V + E, \end{aligned} \tag{28}$$
where $R_e$ is the elevator's capacity per trip (t), and $C_e$ is the elevator launch cost per trip (USD). This accounts for elevator operations and rocket launches.

### 7.2.3 Storage Cost

The unit storage cost $h$ is assumed linear and fixed. Based on NASA cost engineering methods, $h$ is set within a reasonable range of 50 000 USD/t/year to 300 000 USD/t/year, with a nominal value of 120 000 USD/t/year for analysis. The range is used for sensitivity testing.

## 7.3 Constraints

The model includes practical constraints:

$$Q \leq T \times s, \quad \text{(Transportation capacity constraint)} \tag{29}$$
$$Q \leq M, \quad \text{(Storage capacity constraint)} \tag{30}$$

where $T$ is the total transportation capacity per year (t/year), and $M$ is the maximum storage capacity on the Moon (t). The first constraint ensures that the transportation system can deliver $Q$ within time $s$, preventing stockouts due to prolonged shipping.

# 8 Models and Solutions of Problem 4

Sustainable lunar colonization requires careful consideration of environmental impacts alongside traditional economic and operational metrics. This paper extends the inventory-replenishment optimization model for lunar water supply by incorporating environmental impact assessment. We quantify environmental effects primarily through carbon emissions, recognizing rockets as significant emission sources while considering space elevators as near-zero-emission alternatives. The optimization problem becomes tri-objective, minimizing total cost, total time, and total environmental impact simultaneously. We propose three practical strategies for environmental optimization and apply the model to analyze resource requirements for a 100,000-person lunar settlement, comparing them with equivalent terrestrial consumption patterns.

We assume environmental impact per unit mass transported can be estimated as a "damage value," with carbon emissions serving as the primary metric. This allows for standardized comparison across transport modes.

Energy generation at the Galactic Port via solar or nuclear power results in minimal environmental impact (approaching zero), as operational phases occur primarily in space with negligible atmospheric emissions.

Sustainable lunar colonization requires careful consideration of environmental impacts alongside traditional economic and operational metrics. This paper extends the inventory-replenishment optimization model for lunar water supply by incorporating environmental impact assessment. We quantify environmental effects primarily through carbon emissions, recognizing rockets as significant emission sources while considering space elevators as near-zero-emission alternatives. The optimization problem becomes tri-objective, minimizing total cost, total time, and total environmental impact simultaneously. We propose three practical strategies for environmental optimization and apply the model to analyze resource requirements for a 100,000-person lunar settlement, comparing them with equivalent terrestrial consumption patterns.

## 8.1 Model Assumptions

- **Environmental impact quantifiability:** The "environmental damage value" per unit mass (or per launch) for each transportation mode can be estimated using carbon emissions.
- **Galactic Harbour environmental impact:** If powered by solar or nuclear energy, environmental impact is extremely low (approaching zero).
- **Rocket environmental impact** (primarily from launch phase): Includes atmospheric pollution (emissions of $CO_2$, $H_2O$, soot, nitrogen oxides); noise and local pollution affecting ecosystems around launch sites. Given that most pollution can be quantified via carbon emissions, we prioritize focusing on carbon emissions for analysis.
- **Lunar base infrastructure:** Assume the lunar base has waste processing infrastructure; returning waste to Earth is not considered for now.

## 8.2 Data Processing

### Space Elevator Environmental Coefficient

Space elevators, powered by clean energy sources, are assigned a unit environmental impact coefficient $e_e \approx 0$ tons $CO_2$ per ton of cargo.

### Rocket Carbon Emission Coefficient

Based on ISEC 2024 study ("apex anchor" dynamics and operational settings) and authoritative engineering data derivation:

- **Transfer rocket** (for segment transportation from geostationary orbit to the Moon within elevator system): Unit carbon emission coefficient $e_{\text{tran}} = 2.5$ tons $CO_2$/ton cargo.
- **Full-scale rocket** (direct launch from Earth to Moon): Unit carbon emission coefficient $e_{\text{launch}} = 55$ tons $CO_2$/ton cargo.

The derivation process integrates stoichiometry (ESA Clean Space), engine mixture ratio (Merlin technical parameters), mission $\Delta v$ (NASA/ESA guidelines), Tsiolkovsky rocket equation, and is verified against UCL's 3D emission inventory for order-of-magnitude validation.

### Carbon Tax Pricing

For environmental cost internalization, a carbon tax rate $p_c = 50$ USD/ton $CO_2$ is applied (based on U.S. EPA's 2023 environmental damage monetization concept).

### Total Carbon Emission Cap

Set maximum allowable carbon emission cap $B = 3.1$ million tons $CO_2$. This cap is based on IPCC AR6 global remaining carbon budget (approximately 250 $GtCO_2$ for 1.5°C with 50% probability), allocated per capita globally and corresponding to the construction and first-year supply lifecycle limit for the 100,000-person beneficiary population.

## 8.3 Model Formulation

### Environmental Impact Quantification

The total environmental impact $E$ for a transport strategy is:

$$E = \sum_i (m_i \times e_i) \tag{31}$$

where $m_i$ is the mass transported by mode $i$ and $e_i$ is its unit environmental impact coefficient. For a mixed strategy (elevator transport mass $w$, rocket transport mass $Q - w$):

$$E = w \cdot e_e + (Q - w) \cdot e_{\text{launch}} \approx (Q - w) \times 55 \tag{32}$$

### 8.3.1 Multi-Objective Optimization Model

We formulate a tri-objective optimization problem:

**Decision variables:** $\mathbf{x} = (w, s)$, representing mass allocated to elevator and replenishment cycle respectively.

**Total Cost Function** Total cost combines financial and environmental components:

$$C_{\text{total}} = C_{\text{transport}} + C_{\text{storage}} + C_{\text{environmental}} \tag{33}$$

where $C_{\text{transport}}$ is as defined in Section 7.2.2, $C_{\text{storage}} = \frac{Q}{2} \times h$, and $C_{\text{environmental}} = p_c \times E$ if environmental cost is internalized.

**Total Time Function** Total time considers transport duration and cycle management:

$$T_{\text{total}} = \max\left(\frac{w}{R_e}, \frac{Q-w}{R_r}\right) \times \text{missions per cycle} + \text{administrative time} \tag{34}$$

where $R_e$, $R_r$ are elevator and rocket transport rates respectively.

**Environmental Impact Function**

$$E_{\text{total}} = (Q - w) \times e_{\text{launch}} \tag{35}$$

The results are shown as follow:

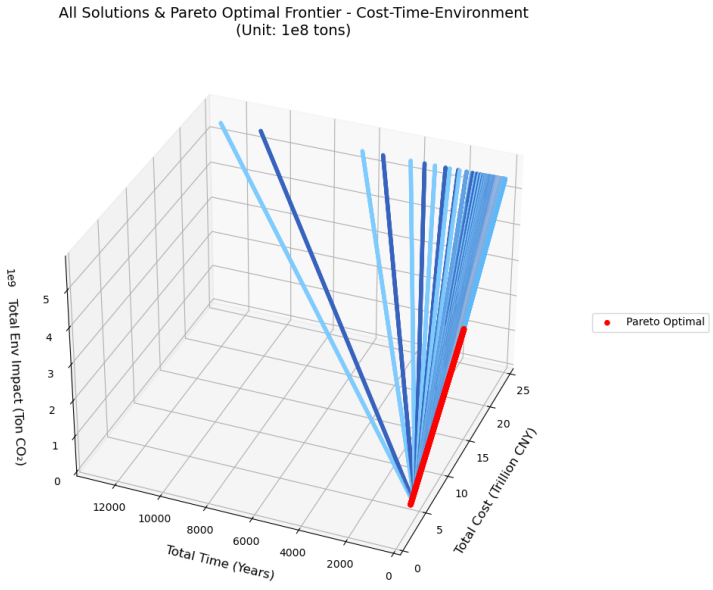



Figure 7: Solutions

## 8.4 Environmental Optimization Strategies

### 8.4.1 Strategy 1: Environmental Damage Monetization

Internalize environmental costs by adding carbon tax to objective:

$$\min \ C_{\text{new}} = C_{\text{financial}} + p_c \times E \tag{36}$$

where $C_{\text{financial}}$ includes only direct transport and storage costs.

### 8.4.2 Strategy 2: Environmental Constraint

Impose carbon emission cap constraint:

$$\min \ C_{\text{total}} \quad \text{subject to} \quad E \leq B \tag{37}$$

converting environmental objective into a constraint.

### 8.4.3 Strategy 3: Clean Transport Priority

Prioritize clean transport capacity, using rockets only when elevator capacity is exhausted:

$$w = \min\left(Q, \ \text{Elevator annual capacity} \times T_{\text{total}}\right) \tag{38}$$

# 9 Sensitivity Analysis

## 9.1 Electricity Price Ratio – Energy Consumption – Maintenance Coefficient

These three parameters are selected as core cost drivers for space elevator operations, which are prone to fluctuations due to technological iterations and energy policies in practice. The analysis results show: In terms of cost, energy efficiency is the most sensitive factor (increasing from 0.2 to 0.8 can significantly reduce costs, and low efficiency amplifies the impact of electricity prices), followed by electricity prices (increases directly drive up transportation costs), while the maintenance coefficient has a weak impact (only causing cost fluctuations of ±5% to +11% even when increasing from 0.1 to 0.4). In terms of time, only energy efficiency is sensitive (transportation time is 72% longer than the baseline when energy efficiency is 0.2). Optimization priority: Improve energy efficiency > Control electricity prices > Optimize maintenance coefficients. When energy efficiency is low (< 0.3), strict control of electricity prices is necessary (cost increases can exceed 50% if it exceeds 0.1 USD/kWh).

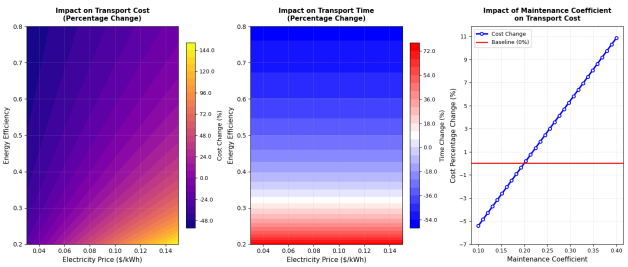



Figure 8

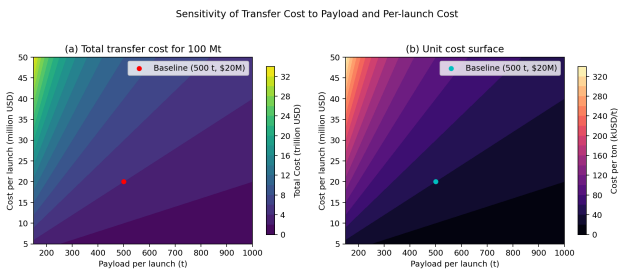



Figure 9

## 9.2 Single Capacity and Single Launch Cost of Transport Rockets

The two are core economic parameters of the rocket transportation mode, strongly coupled to directly determine the total transportation budget, and are significantly affected by rocket technology upgrades and mass production scales in practice. The analysis results: Total expenditure decreases monotonically with the increase of single capacity and the decrease of single launch cost, approximately satisfying $TC \approx M \cdot C/P$; unit ton cost follows the relationship $CPT \approx C/P$, doubling the capacity can halve the ton cost. The target is to control $C/P \leq 10$ kUSD/t, which can constrain the transportation budget of 100 million tons to the trillion-dollar level.

## 9.3 Unit Storage Cost

This parameter directly affects the economy of material reserves at the lunar base. The special lunar storage conditions lead to a wide range of cost fluctuations, and it is also associated with the decision-making of replenishment cycles. The analysis results: Under fixed scenarios (100,000 people, 30 L/person/day, 98% recycling rate, etc.), the annual cost of the "elevator + apex transfer" mode is consistently lower than that of ground-launched rockets, and the costs of both increase linearly with the unit storage cost $h$. The optimal replenishment cycle for the elevator mode is close to 0.5 years (lower bound), so semi-annual replenishment is recommended as the default. The failure probability has a gentle impact on costs, while the delay duration is more sensitive. Efforts should be made to shorten the investigation and re-dispatch time and adopt batch redundancy strategies, and the cost ranking of transportation channels will not reverse.

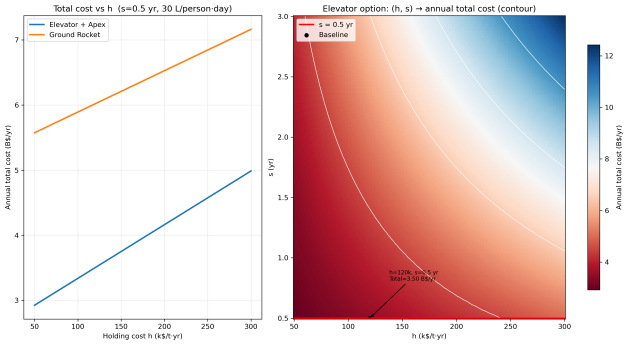



Figure 10

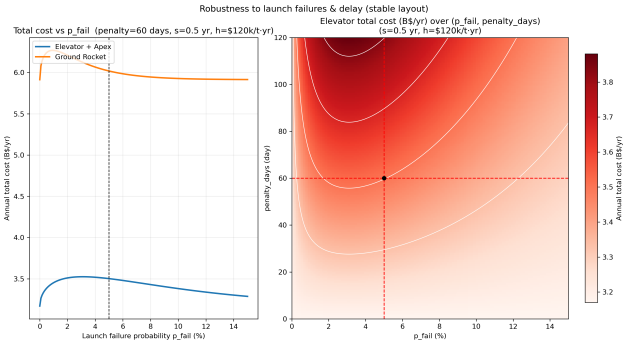



Figure 11

# 10 Strengths and weaknesses

## Strengths

The model has successfully established a multi-objective optimization framework that comprehensively considers cost, time, uncertainty, resource replenishment, and environmental impact. It enables systematic trade-off analysis and provides a scientific basis for decision-making in large-scale Earth-Moon transportation planning.

The model demonstrates good extensibility, from evaluating baseline transport strategies to incorporating risk through Monte Carlo simulation and resource logistics via an EOQ-based inventory

model. This shows that the framework is adaptable to various planning phases and problem scales.

The model exhibits clear interpretability and scenario comparison capability. By presenting results such as Pareto frontiers and strategy tables, it allows stakeholders to visually assess trade-offs between cost, time, and emissions under different operational strategies.

### Weaknesses

Several parameters in the model, such as launch failure rates and carbon emission coefficients, rely heavily on external sources and projected estimates rather than being derived from first principles or real-time lunar mission data. This dependence on external and forward-looking inputs may affect the model's accuracy in actual long-term deployment.

To maintain computational feasibility and clarity in optimization, we simplified certain real-world complexities—such as ignoring launch site geographical differences and assuming independent transport events. While these simplifications make the model more tractable, they may reduce the practical fidelity of the schedule and cost estimates in dynamic, multi-mission scenarios.

## 11 Conclusion

This study addresses each task proposed by the Moon Colony Management (MCM) for the construction and operation of a 100,000-person lunar base, providing targeted optimal solutions as follows: For the transportation of 100 million metric tons of base construction materials across the three scenarios, the hybrid transport strategy is the optimal choice. By rationally combining space elevators and reusable rockets, it achieves a balanced performance in transport efficiency and cost.

In the face of system failures such as tether swaying and rocket launch failures, the risk-adjusted hybrid transport strategy stands out as the most reliable option, ensuring the stable progression of transportation tasks under uncertain conditions.

Regarding the annual water supply needs of 100,000 residents after the base is completed, the EOQ-based hybrid replenishment model is the optimal solution. Through scientific planning of replenishment cycles and transport methods, it guarantees the sustainability of water supply.

Concerning the Earth's environmental impact and mitigation requirements, the tri-objective optimization framework prioritizing clean transport is the best approach, minimizing environmental burdens while meeting transportation demands.

The 100,000-person lunar base also possesses unique value as a "space sustainable resort": Unlike Earth-based resorts that consume ecological resources, it operates with a low environmental footprint relying on efficient resource recycling and in-situ resource utilization. Its distinctive space environment and facility conditions can foster high-end tourism, scientific research, and other diverse industries, generating significant economic benefits while avoiding ecological pressure on Earth, making it a long-term asset with both environmental value and development potential.

# MEMO

**TO:** The Fictional MCM Agency
**FROM:** Team 2631051
**DATE:** February 3, 2026
**SUBJECT:** Quarterly Budget Review

We have completed a thorough analysis of the transportation systems required to deliver materials for the lunar colony. Our study evaluates three approaches: a space elevator system, existing rocket launch facilities, and a hybrid model combining both. The findings are based on mathematical modeling that considers cost, time, reliability, and environmental impact.

Our dynamic transport model incorporates key parameters such as payload capacity, launch frequency, and energy efficiency. For the space elevator system, we accounted for electrical power consumption, maintenance factors, and orbital transfer logistics. For rocket-based transport, we modeled launch sequences, site constraints, and failure probabilities. The results indicate that the space elevator system, while environmentally favorable, has limited annual throughput, extending the project timeline significantly. In contrast, rocket launches offer higher short-term delivery capacity but at substantially greater financial and environmental cost.

We recommend adopting a hybrid strategy that allocates a portion of materials to the space elevator for baseline supply, while utilizing rockets for time-sensitive components. This approach balances cost efficiency with schedule adherence, and can be further optimized using real-time logistics planning.

To support the sustainable operation of the 100,000-person lunar colony post-construction, we have extended our transport model to address annual water supply needs. Focused on a one-year supply cycle, the analysis optimizes inventory replenishment strategies—integrating the hybrid transport framework—to balance transportation costs, lunar storage expenses, and supply reliability. By aligning shipment quantities and replenishment intervals with the capacities of both the space elevator (for steady baseline supply) and rockets (for flexible adjustments), we ensure consistent access to sufficient water without compromising operational efficiency.

Additionally, we have quantified the environmental impact of each transport scenario, with a primary focus on carbon emissions as a key metric. Rocket launches generate significant carbon emissions per ton of cargo, while the space elevator—powered by clean energy—operates with near-zero emissions. To minimize ecological impact on Earth, we have integrated environmental considerations into our optimization framework, prioritizing clean transport capacity and incorporating emission constraints. This approach achieves meaningful reductions in environmental footprint while maintaining manageable trade-offs with cost and timeline objectives, aligning with the long-term sustainability goals of the lunar colony project. Furthermore, we have developed a risk assessment framework that accounts for potential system failures. By integrating reliability metrics and Monte Carlo simulation, we can project the probability of completing transport within budget and timeline constraints.

In summary, a flexible, multi-modal transport plan is essential to the success of the lunar colony project. We stand ready to provide further detail or adapt our models to evolving mission parameters. Thank you for entrusting us with this critical analysis. We look forward to supporting the continued planning and execution of this pioneering endeavor.

# References


[1] International Space Elevator Consortium. (2024). *2024 ISEC study on space elevator dynamics and operations*. Retrieved from https://www.isec.org/2024-study

[2] Luo, S., Cui, N., Wang, X., Fan, Y., & Gu, H. (2022). The stability analysis of a tether for a segmented space elevator. *Aerospace, 9*(7), 376. https://doi.org/10.3390/aerospace9070376

[3] NASA. (2017). *NASA reliability and maintainability (RM) standard for spaceflight and support systems* (NASA-STD-8729.1A). https://s3vi.ndc.nasa.gov/ssri-kb/static/resources/nasa-std-8729.1a.pdf

[4] NASA Kennedy Space Center. (2019, September). *Availability - Concepts and principles for the practitioner*. https://extapps.ksc.nasa.gov/Reliability/Documents/190911.1_Availability_concepts_and_principles_for_the_practitioner.pdf

[5] Intergovernmental Panel on Climate Change. (2023). *AR6 synthesis report: Climate change 2023*. Retrieved from https://www.ipcc.ch/assessment-report/ar6/

[6] U.S. Environmental Protection Agency. (2023). *Report on the social cost of greenhouse gases: EPA's estimates under Executive Order 13990*. Retrieved from https://www.epa.gov/system/files/documents/2023-12/epa_scghg_2023_report_final.pdf

[7] NASA. (2023, December). *Water systems technical brief* (OCHMO-TB-027). Retrieved from https://www.nasa.gov/wp-content/uploads/2023/12/ochmo-tb-027-water.pdf

[8] Wevolver. (n.d.). *Merlin engine (Merlin 1D) technical specifications for Falcon 9 & Falcon Heavy*. Retrieved from https://www.wevolver.com/specs/merlin-engine-merlin-1d-falcon-9-falcon-heavy

[9] Fischer, P. (n.d.). *Life cycle assessment of space transportation systems* [Presentation]. Clean Space Industrial Days, European Space Agency. Retrieved from https://indico.esa.int/event/416/contributions/7288/attachments/4853/7427/CSID_Presentation_Fischer_LC

[10] Japan Aerospace Exploration Agency, Institute of Space and Astronautical Science. (2025). *Microwave Space Solar Power Systems (MSSPS)*. Retrieved February 3, 2026, from https://www.kenkai.jaxa.jp/eng/research/sps/mssps.html

[11] SpaceX. (2025). *SpaceX - Making life multiplanetary*. Retrieved February 3, 2026, from https://www.spacex.com

[12] United Nations Statistics Division. (2021). *Energy statistics yearbook 2021*. Retrieved from https://unstats.un.org/unsd/energystats/pubs/yearbook/2021/t03.pdf

[13] World Health Organization. (n.d.). *Guidelines for drinking-water quality* (4th ed.). Retrieved from https://www.who.int/publications/i/item/9789241549950

Report on Use of AI

1. **Deepseek**
   **Query 1:** Search for relevant literature
   **Output:**

# Literature References: Space Elevators & Lunar Colonies

## Space Elevators

| Category | Key Literature | Author/Publisher | Year |
|---|---|---|---|
| Science Fiction | *The Fountains of Paradise* | Arthur C. Clarke | 1979 |
| | *Red Mars* | Kim Stanley Robinson | 1992 |
| | *Pillar to the Sky* | William R. Forstchen | 2014 |
| Technical Studies | *Space Elevators: An Assessment of the Technological Feasibility and the Way Forward* | International Academy of Astronautics (IAA) | 2013 |
| | *Road to the Space Elevator Era* | International Academy of Astronautics (IAA) | 2019 |

## Lunar Colonies

| Category | Key Literature | Author/Publisher | Year/Notes |
|---|---|---|---|
| Science Fiction | *The Moon Is a Harsh Mistress* | Robert A. Heinlein | 1966 |
| | *Artemis* | Andy Weir | 2017 |
| | *The First Men in the Moon* | H.G. Wells | 1901 |
| Technical Concepts | Colonization of the Moon (Wikipedia Overview) | N/A | — |
| | Academic Study: *"Lunar Colony Novel"* | Huangjiang Diaosuo | 1904 |

2. **Dou Bao (Nov 5, 2023 version, Ernie 4.0)**
   **Query 1:** What are the ways to optimise the chart?
   **Output:** Here are concise ways to optimize charts:

   Simplify design – Remove clutter, unnecessary labels, and decorations.

   Choose the right chart type – Match the chart to your data story.

   Highlight key data – Use color and contrast to draw attention.

Improve readability – Ensure clear labels, legends, and titles.

Use consistent formatting – Maintain uniform colors, fonts, and scales.

Order data logically – Sort by value or categories meaningfully.

Add annotations – Briefly explain important points or trends.

Optimize for viewing platform – Adjust size and detail for screen or print.

Check data accuracy – Ensure correct representation without distortion.

Test with audience – Verify clarity and effectiveness with others.

3. **GitHub Copilot (Feb 3, 2026 version)**
   **Query 1:** Help me correct grammar and spelling mistakes
   **Output:** Main modification instructions:

   The punctuation and format have been unified, such as changing "$CO_2$" to "CO2" and standardising the use of dashes.

   Some sentence structures have been adjusted to make them smoother, such as deleting the spaces in "which we adopt in our failure modelling [3, 4]".

   Minor grammar and spelling problems have been corrected to ensure the rigour and consistency of academic expression.

   Maintain the technical terms, citation format and academic style of the original text.

   If you need to further adjust the tone or style, please let us know at any time.